

NAM TRAN NGUYEN

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EDUCATION

University of Central Florida	Ph.D. in Electrical Engineering, GPA: 4.0/4.0	2025 – Now
Northern Arizona University	M.S. in Electrical Engineering, GPA: 4.0/4.0	2024 – 2025
Hanoi Univ. of Science & Technology	B.S. in Automation & Control Eng., GPA: 3.66/4.0	2019 – 2023

EXPERIENCE

Graduate Research Assistant - University of Central Florida Florida, USA

Uncertainty-Aware Physics-Informed Machine Learning for Robotic Control 05/2026 – Now

- Targeting unreliable predictions of **hybrid physics-informed ML** models for robots in uncertain environments, where the learned part is better in-distribution and the physics part is better **out-of-distribution**.
- Developed an add-on **uncertainty quantification** module for a pre-trained hybrid model that estimates its epistemic uncertainty and uses it to set how much physics and learning each contribute to the prediction.
- Outperformed four baselines in **model predictive control** of a racing car, quadrotor, and Franka manipulator, cutting quadrotor position error **21%** (no disturbance) and **13%** (high noise) versus the hybrid baseline.

Online System Modeling and Adaptive Control for Autonomous Racing 10/2025 – 05/2026

- Addressed real-time **trajectory tracking** under unknown, time-varying road-surface conditions, where fixed-model controllers lose accuracy and the **sim-to-real** gap degrades driving performance.
- Proposed an **online system modeling** method for tire-road interaction that jointly estimates vehicle states and tire parameters in real time, adapting the model to changing road surfaces.
- Designed a **differentiable MPC** on a physics-informed vehicle model that optimizes the controller's cost weights, plus a learned approximation of its solutions for weight tuning.
- Validated in a full-size race-car simulator with nonlinear and linearized solvers under hard real-time constraints, reaching **15% faster** lap times than standard MPC.

Data-Efficient System Identification for Dynamical Systems 05/2025 – 10/2025

- Investigated **optimal experiment design** for identifying dynamical systems under operational constraints, selecting the most informative samples through online **uncertainty quantification** to cut data needs.
- Benchmarked **14** active-learning strategies across neural and probabilistic models on a high-fidelity building simulator, reducing RMSE by up to **54%** versus passive random sampling.

Graduate Research Assistant - Northern Arizona University Arizona, USA

Learning-Based System Modeling for Dynamical Systems 01/2024 – 05/2025

- Compared physics-informed, physics-agnostic, and grey-box surrogate models on real-world time series from building systems, analyzing multi-step prediction error and the failure modes of each approach.
- Introduced a **physics-constrained** network for multi-step prediction with **25% higher** accuracy than standard ML; monotonicity and boundedness constraints on another model gave **50% higher** accuracy than vanilla ML.

Embedded Learning-Based Control on Microcontroller Hardware 01/2024 – 05/2025

- Integrated a physics-informed surrogate into a learning-based **MPC** controller for a **MIMO** system.
- Deployed the controller on **Arduino** hardware, **wiring sensors and actuators** into a closed loop and holding steady-state error below **2%** under disturbances.

Real-Time Vision on NVIDIA Edge Hardware 01/2024 – 05/2024

- Implemented real-time face-recognition pipelines with deep-learning object detectors and deployed them on **NVIDIA Jetson** edge hardware, reaching real-time inference from an **RGB-D** camera.

- Engineered **fuzzy-based** traction-control algorithms for EV acceleration to curb excessive wheel slip and improve drivetrain efficiency and safety.
- Verified closed-loop controllers in a **hardware-in-the-loop** environment using **dSPACE** and TI microcontrollers, cutting wheel slip ratio by **23%** during acceleration.

TECHNICAL SKILLS

- **Estimation & Modeling:** MHE, EKF/UKF, Particle Filters, System Identification, Physics-Informed ML
- **Control & Learning:** Model Predictive Control, Reinforcement Learning, Uncertainty Quantification
- **Programming:** Python (PyTorch, JAX, CasADi, NumPy, SciPy), C/C++, MATLAB/Simulink, ROS2
- **Tools & Simulation:** MuJoCo, F1TENTH-Gym, IPOPT/OSQP, Linux, Git, Docker, HPC
- **Hardware:** NVIDIA Jetson, Raspberry Pi, Arduino Uno, dSPACE R&D Board, TI Microcontrollers

PUBLICATIONS

- **Nam T. Nguyen**, Binh Nguyen, Ahmad Amine, et al., "AD-MPCC: Adaptive Differentiable Model Predictive Contouring Control for Autonomous Racing Car," 2026. (Accepted to IROS 2026) Link: arxiv.org/abs/2607.00141
- **Nam T. Nguyen**, J. C. Tique, and Truong X. Nghiem, "Physics-Constrained Taylor Neural Networks for Learning and Control of Dynamical Systems," *2025 American Control Conference*, Denver, Colorado, USA.
- Binh Nguyen, **Nam T. Nguyen**, and Truong X. Nghiem, "Structure- and Stability-Preserving Learning of Port-Hamiltonian Systems," 2026. (Accepted to CDC 2026) Link: arxiv.org/abs/2604.13297
- **Nam T. Nguyen**, Truong X. Nghiem, Minh N. Vu, "EPIC-PIML: Epistemic-Constructed Physics-Informed Machine Learning." (In preparation for ICLR 2027)
- **Nam T. Nguyen**, Binh Nguyen, and Truong X. Nghiem, "Physics-Informed Data-Driven Modeling of HVAC Systems: A Systematic Analysis," *IEEE Access*, 2026.
- **Nam T. Nguyen** and Truong X. Nghiem, "Active Learning for Optimal Experimental Design in Machine Learning-Based Building Energy System Identification," 2026. (Submitted to Energy and Buildings)
- **Nam T. Nguyen**, Minh C. Ta, Thanh Vo-Duy, and Valentin Ivanov, "Enhanced Fuzzy-MFC-based Traction Control System for Electric Vehicles," *IEEE Vehicular Power and Propulsion Conference 2023*, Milan, Italy.

AWARDS

- Member of Tau Beta Pi Engineering Honor Society, Florida Delta Chapter (UCF), 2026
- UCF College of Graduate Studies Presentation Fellowship 2026
- UCF Open Access Publishing funds 2026
- Travel Award, American Control Conference (ACC), 2025
- Support for Graduate Students (SGS) Summer Research Award, Northern Arizona University, 2024
- LG Display Scholarship for Outstanding Students, 2022
- Excellent Study Scholarship (Top 1% of students), Hanoi University of Science and Technology, 2020, 2021, 2023
- Best Presentation Award, Student Forum on Sustainable Energy, 2021
- Best Presentation Award, Student Forum on Frontiers in Sustainability, 2023

LEADERSHIP

- Founder and host of the Research Talk series for Vietnamese engineering researchers around the world with 6,600 subscribers and 30,000 watch time hours: [@NamNguyen.ee2](https://www.youtube.com/@NamNguyen.ee2).
- Leader of Vietnamese international student community at the University of Central Florida, USA.